FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 1.4: ENERGY, WORK, POWER AND MACHINES	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

You have spent 8 lessons investigating why a jua kali mechanic in Gikomba can loosen a stubborn matatu wheel nut effortlessly with a long spanner extension — and why the nut gets warm after repeated failed attempts with a short one. Now it is time to bring all your evidence together into one well-reasoned Final Explanation.

Your Final Explanation must directly answer BOTH parts of the driving question:
(1) Why does a longer spanner make loosening the nut easier?
(2) Where does the energy go when friction makes the nut hot?

Use the section prompts below to build your answer. In each section, refer back to specific evidence you gathered — from the lever experiment, the work-energy calculations, the friction investigation, and the simple-machines models. Write in full sentences and use correct Physics vocabulary throughout. Support every claim with data or reasoning, not just definitions.

KICD Competencies you are demonstrating: Critical thinking and problem-solving; Communication and collaboration; Learning to learn.
KICD Core Values: Responsibility, Integrity, and Respect for evidence.
Pertinent and Contemporary Issues (PCIs): Energy conservation in everyday Kenyan workplaces; Safety for jua kali workers.

Section 1 — Mechanical Advantage: Why the Long Spanner Works	
Explain, using the law of the lever and the concept of mechanical advantage (MA), why the jua kali mechanic succeeds with the long spanner extension but fails with the short spanner. Your answer must: (a) state the lever principle (moment = force ×	A spanner works as a Class 1 lever, where the pivot is the centre of the wheel nut. The turning effect — called the moment or torque — is calculated as: Moment (Nm) = Force (N) × perpendicular distance from pivot (m). When the mechanic uses a short 0.15 m spanner and pushes with 200 N, the moment is only 200 × 0.15 = 30 Nm — not enough to overcome the nut's resistance. By sliding a metal pipe over the spanner to make the effort arm 0.60 m, the same 200 N push now produces 200 × 0.60 = 120 Nm — four times larger — which easily exceeds the nut's resistance. The mechanical advantage (MA) of a lever = Load Force ÷ Effort Force. With the short spanner, if 500 N is needed to break the nut free but only 30 Nm is produced, you would need to push with 500 N (MA ≈ 1). With the long extension, you only need to push with about 125 N to produce the same 75 Nm needed (MA = 500 ÷ 125 = 4). This means the long spanner multiplies your force by a factor of 4 — the 'extra' force does not appear from nowhere; it comes from the trade-off of pushing over a longer distance. The mechanic's muscles do the same amount of work, but the machine

perpendicular distance) and identify the effort arm and load arm in the spanner system; (b) show, using numbers from your investigation, how increasing the effort arm increases the moment applied to the nut; (c) define mechanical advantage and calculate it for at least two different spanner lengths you tested; and (d) explain why a greater MA means the mechanic needs to apply less muscular force to produce the same turning effect on the nut.	redistributes the force over distance, making the effort feel smaller.

Section 2 — Work, Energy, and the Conservation Principle	
Explain how the principle of conservation of energy applies to the spanner-lever system. Your answer must: (a) define work done (W = F × d) and explain the work-energy theorem in your own words; (b) show why a simple machine like a spanner cannot create extra energy — use the concept of 'work input = work output + energy lost to friction' to support your answer; (c) use a worked example with realistic numbers (effort force, effort distance, load force, load distance) to demonstrate that work input ≈ work output when friction is low; and (d) connect this back to the mechanic's situation — why does making the spanner longer NOT give the mechanic 'free' extra energy?	Work is done whenever a force moves an object through a distance in the direction of the force: W = F × d (measured in Joules). The work-energy theorem tells us that the net work done on an object equals the change in its kinetic energy. For machines, the key principle is conservation of energy: energy cannot be created or destroyed, only transferred or transformed. This means a spanner cannot manufacture extra energy — it can only redistribute what the mechanic's muscles supply. Consider the long spanner (effort arm = 0.60 m, load arm = 0.05 m, the distance from nut centre to spanner jaw). If the mechanic pushes with 125 N through an arc of 0.30 m at the handle end, work input = 125 × 0.30 = 37.5 J. The nut (the load) rotates through a much smaller arc — about 0.025 m at the jaw end — but experiences a much larger force of approximately 500 N. Work output = 500 × 0.025 = 12.5 J... wait — that is less than the input! The difference (37.5 − 12.5 = 25 J) is lost to friction between the spanner jaw and nut faces, and between the nut and bolt threads. In an ideal (frictionless) machine: work input = work output. In reality: work input = useful work output + energy dissipated as heat. So the long spanner does NOT give the mechanic free energy; it simply lets him apply a smaller force over a larger distance to achieve the same — or greater — turning effect. The 'effortless' feeling is a redistribution of force and distance, not a creation of energy.

Section 3 — Energy Dissipation: Why the Nut Gets Hot

Explain what happens to the energy that seems to 'disappear' when the apprentice struggles with the short spanner and the nut becomes warm. Your answer must: (a) describe what friction is at a molecular/surface level and why it acts between the spanner jaw, nut faces, and bolt threads; (b) explain the energy transformation chain: chemical energy (muscles) → kinetic energy (spanner movement) → thermal energy (heat in the nut and spanner); (c) use the first law of thermodynamics / conservation of energy to argue that the energy has NOT disappeared but has been transformed; and (d) explain why this thermal energy is 'wasted' from the mechanic's point of view — it cannot easily be recovered to do useful mechanical work on the nut.	When the apprentice pushes and twists with the short spanner repeatedly without success, he is still doing work — but almost none of it goes into rotating the nut. Instead, the microscopic rough surfaces of the spanner jaw and nut grind against each other. Friction occurs because even surfaces that look smooth are covered in tiny bumps and valleys at the molecular scale; when two surfaces slide, these bumps interlock and resist motion, and the energy needed to break those interlocking bonds is released as thermal energy (heat). The energy transformation chain is: Chemical potential energy stored in the apprentice's muscles → kinetic energy of the moving spanner → thermal energy (heat) in the nut, spanner, and bolt threads. This is why the nut feels warm. According to the first law of thermodynamics (a form of energy conservation), the total energy of the system is unchanged: energy input from muscles = useful work done rotating the nut + heat generated by friction. The energy has NOT disappeared — it has transformed into thermal energy, raising the temperature of the nut and surrounding metal. However, from the mechanic's point of view this thermal energy is 'wasted' because: (1) heat spreads out in all directions into the metal and air and cannot easily be collected back; (2) it does not contribute to the useful task of rotating the nut. This is why real machines always have an efficiency less than 100% — some input energy is always dissipated as heat due to friction, and cannot be recovered as useful mechanical work.

Section 4 — Efficiency, Power, and Real-World Machines	
Connect mechanical advantage, energy conservation, and friction by discussing the efficiency and power of the spanner system. Your answer must: (a) define efficiency (%) = (useful work output ÷ total work input) × 100 and explain what reduces it in the spanner system; (b) define power (P = W ÷ t) and explain why power matters to the jua kali mechanic — for example, why does working faster or slower change the experience even if the total work done is the same? (c) suggest TWO	Efficiency measures how well a machine converts work input into useful work output: Efficiency (%) = (Useful Work Output ÷ Total Work Input) × 100. In the spanner system, efficiency is reduced mainly by friction between the spanner jaw and nut (energy lost as heat) and by any bending or flexing of the pipe extension (energy stored as elastic deformation). If work input = 37.5 J and useful work output = 12.5 J, then efficiency = (12.5 ÷ 37.5) × 100 = 33.3% — meaning two-thirds of the mechanic's energy is wasted as heat! Power is the rate of doing work: P = W ÷ t (measured in Watts). Even if the mechanic does the same total work, doing it faster requires greater power output from his muscles. A mechanic who loosens the nut in 2 seconds exerts P = 37.5 ÷ 2 = 18.75 W, while one who takes 10 seconds exerts only 3.75 W. Higher power demands tire muscles faster, which is why working steadily with the long spanner (lower force, manageable power) is less exhausting than struggling rapidly with the short one. To improve efficiency: (1) Apply grease or oil (lubricant) to the nut threads — lubricant fills the microscopic surface gaps, reducing direct metal-to-metal contact and therefore reducing frictional heat losses, so more input energy goes into useful rotation; (2) Use a well-fitting spanner jaw — a loose jaw rocks on the nut, increasing contact surface irregularity and friction losses, while a snug-fitting jaw transfers force more directly. In Kenyan jua kali workshops and larger industries, improving machine efficiency through lubrication and proper tool maintenance reduces energy waste, cuts fuel and electricity costs, extends tool life, and protects workers from heat-related injuries.

practical ways the mechanic could improve the efficiency of his spanner system (e.g., lubrication, cleaner spanner jaw fit) and explain the Physics behind each suggestion; and (d) relate your answer to the broader principle: in Kenyan workplaces and industries, improving machine efficiency means less energy wasted, lower fuel costs, and safer working conditions for jua kali workers.	

Section 5 — Answering the Driving Question	
Now bring everything together to write a complete, direct answer to the driving question: 'Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot?' Your answer must: (a) directly answer BOTH parts of the driving question in a clear opening statement; (b) reference at least THREE Physics concepts from this unit (e.g., moment/torque, mechanical advantage, work, conservation of energy, efficiency, power, friction, energy transformation); (c) connect your answer back to specific evidence or data from your investigations; and (d) reflect on what this means for real Kenyan workers — how does understanding these principles help jua kali mechanics, engineers, and policymakers design better	The jua kali mechanic can loosen the wheel nut effortlessly with a long spanner because increasing the length of the effort arm dramatically increases the moment (turning force) applied to the nut — this is mechanical advantage at work. The energy that 'disappears' during the struggle with the short spanner has not vanished; it has been transformed into thermal energy (heat) by friction, exactly as the law of conservation of energy predicts. Together, these two ideas — mechanical advantage and energy transformation — fully explain the phenomenon we observed in Gikomba. In detail: the spanner is a lever with its pivot at the centre of the wheel nut. Moment = Force × perpendicular distance, so doubling the spanner arm from 0.15 m to 0.60 m quadruples the turning moment for the same muscular effort, raising mechanical advantage from 1 to 4. This is not 'free' force — it is a trade-off: the mechanic's hand moves through four times the distance, doing the same total work (W = F × d) but with one-quarter of the force. Conservation of energy is not violated: work input = useful work output + energy lost to friction. When the apprentice struggles with the short spanner, nearly all his work input is converted to heat through friction between metal surfaces — this raises the nut's temperature, which is exactly what he observes. The efficiency of the system (useful output ÷ total input × 100%) is far below 100% because of these frictional losses, and the wasted thermal energy disperses into the surroundings and cannot be recovered. For Kenyan workers, understanding these principles is powerful: a jua kali mechanic who knows about moments will choose the right tool length and save time and energy. An engineer designing matatu maintenance equipment can specify longer torque wrenches and better lubrication systems to reduce worker fatigue and heat injuries. Policymakers supporting Kenya's jua kali sector can invest in quality, well-fitting tools and training programmes — because a mechanic who understands the Physics of their tools works more safely, more efficiently, and with less wasted energy, contributing to a more productive and sustainable Kenyan economy.

tools and safer, more energy-efficient workplaces in Kenya?	

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Mechanical Advantage & Lever Principle	Accurately explains the lever principle (moment = F × d) and correctly identifies effort arm and load arm in the spanner system. Calculates mechanical advantage for at least two spanner lengths using correct data from investigations. Clearly explains the force-distance trade-off with no misconceptions.	Correctly states the lever principle and calculates mechanical advantage for at least one spanner length. Explains the trade-off between force and distance with minor gaps or imprecision in language.	States that a longer spanner helps but cannot correctly apply moment = F × d or calculate mechanical advantage. Explanation of the force-distance trade-off is missing or contains significant errors.
Conservation of Energy & Work	Precisely defines work (W = F × d) and states the conservation of energy principle. Uses a worked numerical example to show work input = useful work output + heat lost to friction. Clearly explains why the machine creates no extra energy. All values are consistent and correctly labelled with units.	Defines work correctly and states conservation of energy. Provides a mostly correct numerical example with minor arithmetic or unit errors. Explains that energy is not created by the machine but may lack full clarity on the friction loss term.	Defines work or energy conservation in general terms only, without applying them numerically to the spanner system. May claim the longer spanner 'creates' extra force or energy, showing a misconception about conservation.
Energy Dissipation & Friction	Explains friction at a surface/molecular level and traces the full energy transformation chain (chemical → kinetic → thermal). Uses conservation of energy to argue convincingly that heat energy is not lost but transformed. Clearly distinguishes 'useful' from 'wasted' energy and explains why thermal energy cannot be easily recovered.	Describes friction and the energy transformation chain correctly at a general level. States that energy is conserved and transformed to heat, but the molecular-level explanation or the reasoning about recoverability of heat is incomplete.	States that friction produces heat but cannot explain the energy transformation chain or why energy is not truly 'lost.' May describe energy as disappearing rather than transforming, indicating a misconception about conservation.
Efficiency & Power	Correctly defines and calculates efficiency (%) using values from the investigation. Defines power (P = W ÷ t) and explains its relevance to the mechanic's experience. Proposes two well-explained, Physics-grounded strategies to improve efficiency (e.g., lubrication, proper jaw fit). Connects improved efficiency to real benefits for Kenyan jua kali workers.	Correctly defines efficiency and power. Calculates efficiency with minor errors. Proposes at least one strategy to improve efficiency with a reasonable Physics explanation. Makes a general connection to Kenyan workplace relevance.	Defines efficiency or power with significant errors, or confuses the two concepts. Cannot calculate efficiency from given data. Strategies to improve efficiency are vague or not connected to Physics principles.

Synthesis: Answering the Driving Question	Opening statement directly and completely answers both parts of the driving question. Integrates at least three Physics concepts from the unit, referenced with specific evidence or data. Reflection on real-world implications for Kenyan jua kali workers, engineers, or policymakers is insightful, specific, and scientifically grounded. Writing is clear, well-organised, and uses correct Physics vocabulary throughout.	Answers both parts of the driving question but one part may lack depth or specificity. References at least two Physics concepts with some connection to evidence. Includes a relevant but general reflection on Kenyan workplace implications. Writing is mostly clear with occasional vocabulary or organisation issues.	Attempts to answer the driving question but addresses only one part, or gives a superficial answer to both. Physics concepts are named but not meaningfully connected to evidence or the phenomenon. Reflection on real-world implications is absent or not connected to the Physics. Writing is unclear or disorganised.